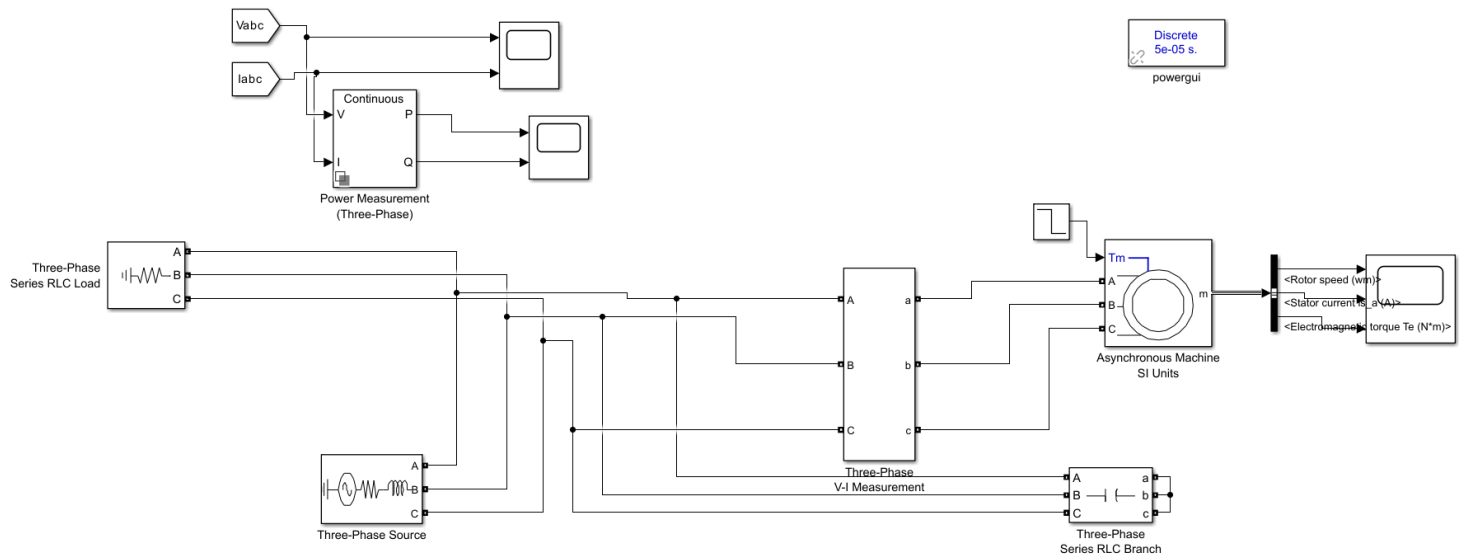


Exp2 SEIG



Block Parameters: Three-Phase Source

Three-Phase Source (mask) (link)

Three-phase voltage source in series with RL branch.

Parameters Load Flow

Configuration: Yg

Source

☐ Specify internal voltages for each phase

Phase-to-phase voltage (Vrms): 400

Phase angle of phase A (degrees): 0

Frequency (Hz): 50

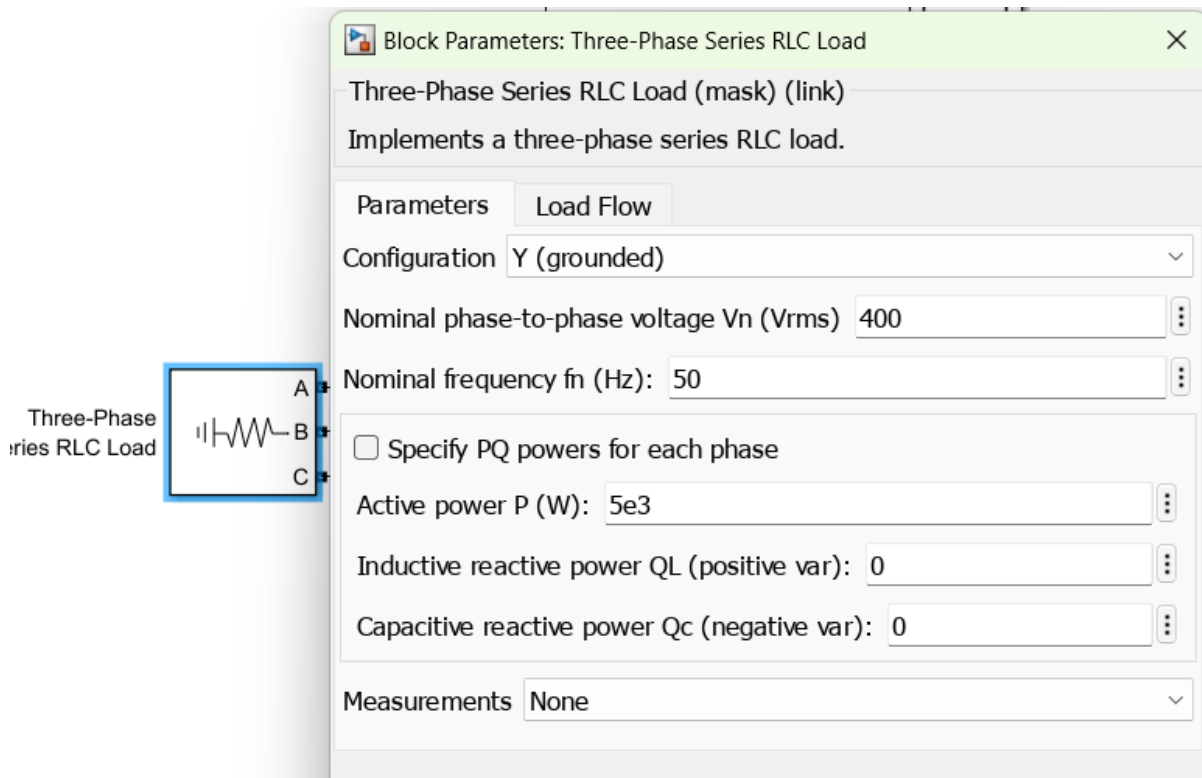
Impedance

☒ Internal ☒ Specify short-circuit level parameter

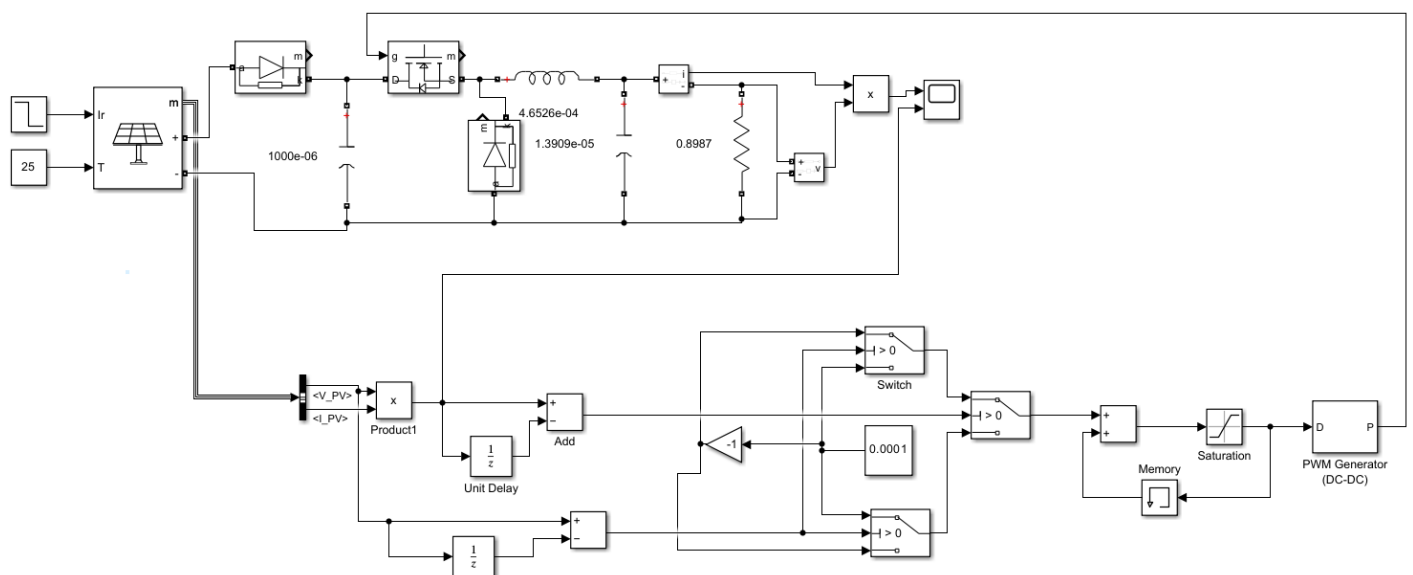
3-phase short-circuit level at base voltage(VA): 10e3

Base voltage (Vrms ph-ph): 400

X/R ratio: 7



Exp3: MPPT alg



Exp7: IV PV

```
clc;
clear;

Voc = 30;      % Open circuit voltage (V)32.9
Isc = 8;       % Short circuit current (A)8.21

V = 0:0.1:Voc;

% Simple approximate I-V curve
I = Isc * (1 - (V/Voc).^2);

% Power
P = V .* I;

% I-V curve
figure;
plot(V,I,'LineWidth',2);
xlabel('Voltage (V)');
ylabel('Current (A)');
title('PV I-V Characteristics');
grid on;

% P-V curve
figure;
plot(V,P,'LineWidth',2);
xlabel('Voltage (V)');
ylabel('Power (W)');
title('PV P-V Characteristics');
grid on;
```

with knee

```
clc;
clear;
close all;

Voc = 32.9;      % Open circuit voltage (V)32.9
Isc = 8.21;      % Short circuit current (A)8.21

V = 0:0.1:Voc;

% PV I-V characteristic
Vknee = 24;      % Knee voltage
I = Isc * (1 - (V/Vknee).^10);

% Set current to zero after knee
I(V > Voc) = 0;
I(I < 0) = 0;

% Power
P = V .* I;

% I-V characteristic
figure;
plot(V,I,'LineWidth',2);
```

```

xlabel('Voltage (V)');
ylabel('Current (A)');
title('PV I-V Characteristics');
grid on;

% P-V characteristic
figure;
plot(V,P,'LineWidth',2);
xlabel('Voltage (V)');
ylabel('Power (W)');
title('PV P-V Characteristics');
grid on;

```

exp6 : energy storage system

```

clc;
clear;

% Battery parameters
Capacity = 936;           % Battery capacity
SOC = 0.2;                % Initial SOC = 20%
I = 5;                    % Charging current (A)
Vnom = 12;                % Nominal battery voltage (V)

% Time
t = 0:0.1:80;

% Calculate SOC
SOC = SOC + (I * t) / Capacity;

% Limit SOC to 100%
SOC(SOC > 1) = 1;

% Simple battery voltage model
Vbat = Vnom + 2*SOC;

% Plot battery voltage
figure;
plot(t,Vbat,'LineWidth',2);
xlabel('Time (s)');
ylabel('Battery Voltage (V)');
title('Battery Voltage vs Time');
grid on;

% Plot SOC
figure;
plot(t,SOC,'LineWidth',2);
xlabel('Time (s)');
ylabel('State of Charge');
title('SOC vs Time');
grid on;

```

fuel cell:

```
clc;
I = 0.1:0.1:10;
T = 300;
A = 70;

z1 = -0.5;
z2=0.005;
z3 = 7e-7;
z4 = -7e-7;
co2=7e-7

Rm = 0.2;
B = 0.02;
Rc = 0.02;

Jmax = 1600;
N = 24;

% Current density
J = I/A;

% Voltage losses
Vact = -(z1 + z2*T + z3*T*log(co2) + z4*T*log(I));
Vohm = I*(Rm + Rc);
Vcon = -B*log(1 - J/Jmax);

% Fuel cell voltage
EN=3;
Vfc = N*(EN - Vact - Vohm - Vcon);

% Fuel cell power
P = Vfc .* I;
% Plot
yyaxis left
plot(I,Vfc,'LineWidth',1.5);
ylabel('Fuel Cell Voltage (V)');
xlabel('Fuel Cell Current (A)');

yyaxis right
plot(I,P,'LineWidth',1.5);
ylabel('Fuel Cell Power (W)');
```

pviv practically easy

```

clc;
clear;
Rp=400;
Voc=30;
a=2;
I0=5e-05;
Ipv=8;
Vt=1.2;
V=0:0.1:Voc;
I=Ipv-I0.*(exp(V/(a*Vt))-1)-V/Rp;
I(I<0)=0;

```

```

P=V.*I;

```

```

figure;
plot(V,I,'LineWidth',2)
xlabel('Voltage (V)')
ylabel('Current (A)')
title('I-V Characteristics')
grid on

```

```

figure;
plot(V,P,'LineWidth',2)
xlabel('Voltage (V)')
ylabel('Power (W)')
title('P-V Characteristics')
grid on

```

exp1 pmsm

